

Alex Junior Gomez Saltachin

PhD student in Mathematics, PUC-Rio

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Research Profile

I am a PhD student in Mathematics at PUC-Rio, supervised by Sergey Galkin with Vasily Golyshev as co-advisor. My work lies at the interface of algebraic geometry, birational geometry, Homological Mirror Symmetry, derived categories, Fano and log del Pezzo geometry, Landau-Ginzburg models, periods, monodromy, and arithmetic aspects of mirror symmetry.

Current project: big quantum cohomology for log del Pezzo surfaces, with X_{10} in $P(1,2,3,5)$ as a central example.

Education

- **PhD in Mathematics**, PUC-Rio, Brazil, 2023-present. Advisor: Sergey Galkin. Co-advisor: Vasily Golyshev.
- **MSc in Mathematics**, Pontificia Universidad Catolica del Peru, 2019. Thesis: *Modular Forms*.
- **BSc in Mathematics**, Pontificia Universidad Catolica del Peru, 2015. First Class Honors. Thesis: *Elliptic Curves: Mordell-Weil, Roth and Siegel Theorems*.

Honors and Fellowships

- **FAPERJ Nota 10 Excellence Scholarship**, Rio de Janeiro, Brazil, 2024-present.
- First Class Honors, BSc in Mathematics, Pontificia Universidad Catolica del Peru, 2015.

Preprints, Manuscripts, and Drafts

- **Exceptional Collections for Canonical Stacks of Log del Pezzo Surfaces with $1/3(1,1)$ -Singularities**. arXiv:2606.18238, posted June 16, 2026; submitted to *Advances in Geometry* at De Gruyter.
- Summary: proves full exceptional collections for canonical stacks of complex log del Pezzo surfaces with $1/3(1,1)$ -singularities, obtains the length formula $\ell = 12 - K_X^2 + 4r/3$, treats the six qG-rigid cascade bases explicitly, and analyzes X_{10} in $P(1,2,3,5)$, KKS descent, and Brauer groups.
- **A Seven-Parameter Landau-Ginzburg Mirror for the Cubic Surface**, in preparation.
- **Categorical Mirror Genus for Orbifold del Pezzo Surfaces and the Genus-Two Mirror of X_{10}** , in preparation.
- **Arithmetic Progressions in Gap Sets of Numerical Semigroups**, draft in preparation; computational certificates organized on GitHub.
- **Real Bruhat Intersections and Complex Open Richardson Varieties**, draft communication, 2026.

Selected Talks and Posters

- *Brauer-Manin and a log del Pezzo surface*, Pontificia Universidad Catolica de Chile, 2026.
- *Exceptional Collections for Canonical Stacks of Log del Pezzo Surfaces with $1/3(1,1)$ -Singularities*, poster, RGAS Summer School, Zaragoza, Spain, 2026.
- *Periods in Homological Mirror Symmetry and Kloosterman Motives*, ICTP, 2025.
- *A Tentative Relationship Between Homological Mirror Symmetry and Arithmetic Geometry*, Arithmetic and p-adic Geometry in Chile, 2024.
- *The Mordell-Weil Theorem*, PUC-Rio Friday Seminar, 2024.
- *Karoubian Completion of Triangulated Categories*, PUC-Rio Friday Seminar, 2024.
- *Model structures*, PUC-Rio Friday Seminar, 2024.
- *A real analogue of the Moore-Tachikawa category*, PUC-Rio Friday Seminar, 2024.
- *Delzant's theorem on toric symplectic manifolds*, PUC-Rio Friday Seminar, 2023.

Teaching and Notes

- Graduate instructor and teaching assistant at PUC-Rio and PUCP, 2013-2026.
- Courses taught or assisted include Linear Algebra, Algebra, Real Analysis, Complex Analysis, Calculus, and Mathematics for Economists.
- Notes on quotient groups and algebraic structures; derived categories of sheaves; and Geometric Invariant Theory.

Academic Service

- Organizer of the Etale Cohomology Seminar 2026, a five-week online reading and discussion seminar following Daniel Litt's course *Etale Cohomology and the Weil Conjectures*, starting August 28, 2026.
- One of the typesetters, with Mohamed Aliouane, for Masha Vlasenko's lecture notes *Hypergeometric Functions* in the ICTP Number Theory and Physics Proceedings, in progress.
- Teaching assistant for *Metodos explicitos de Chabauty: un punto de vista computacional a los puntos racionales*, AGRA V, Popayan, Colombia, 2026.
- Co-founder and organizer of the 38 Movimientos Seminar.
- Co-organizer of the Meetings on Arithmetic and Algebraic Number Theory Seminar.

Selected Schools and Programs

- Rutgers Symplectic Summer School, Rutgers University, New Brunswick, USA, 2026.
- AGRA V: School on Arithmetic, Groups and Analysis, Popayan, Colombia, 2026.
- GARLAT: Latin American School in Arithmetic Geometry, Santiago, Chile, 2026.
- RGAS Summer School: Mirror Symmetry and Homological Methods in Algebraic Geometry, Zaragoza, Spain, 2026.
- ICTP School and Workshop on Explicit Arithmetic Geometry, Trieste, Italy, 2025.
- The Mordell Conjecture 100 Years Later, MIT, USA, 2024.
- ICTP String-Math, Trieste, Italy, 2024.
- CIMPA ELGA - Latin American School of Algebraic Geometry, 2024.
- Arizona Winter School, 2022 and 2023; IHES Summer School on the Langlands Program, 2022.
- ICTP-CIMPA Schools in Arithmetic Geometry, 2015 and 2018.

Languages

Spanish native; English advanced; Portuguese advanced; basic Chinese and French.